Jacobian transfer discovery across the curve registry

automated screen, 2026-10-10

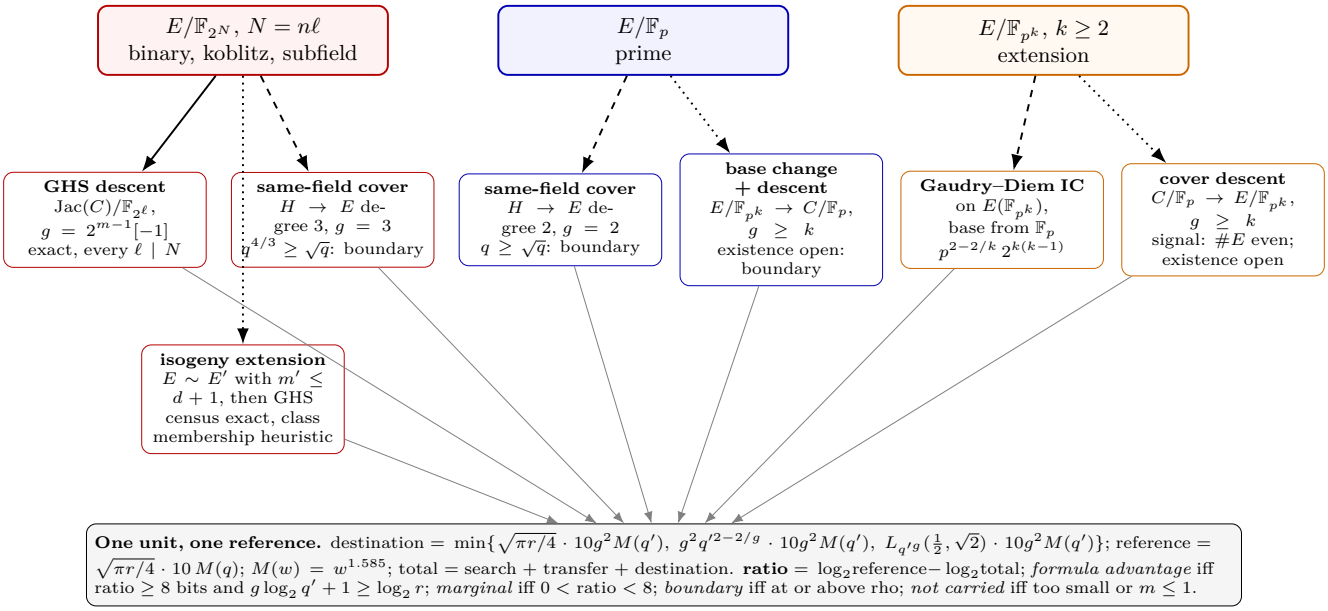
crypto repository, `jacobian_transfers`

2026-10-10

Question and answer

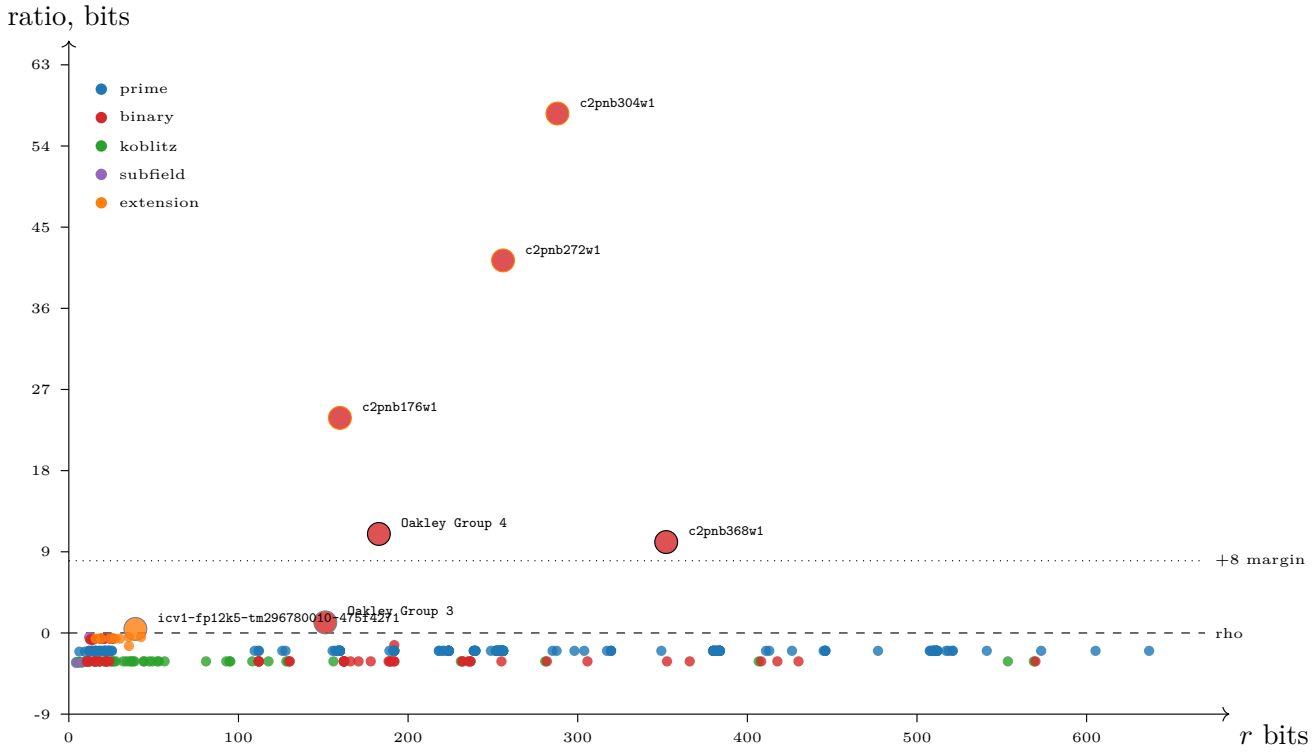
For each of the **321** curves in `docs/curves/registry.json`, can its discrete logarithm be moved into the Jacobian of a curve of higher genus at a cost the literature's complexity formulas place below Pollard rho on the original subgroup? The native screen (`src/cryptanalysis/jacobian_transfer/`) enumerates the known routes, computes what is exact (GHS magic numbers and genera for every field factorisation, same-field cover genera and degrees, 2-torsion signals, the census of curves with small Frobenius annihilators), and prices the rest with one frozen formula cost model. Every row is a formula. A *formula advantage* is a lead with open obligations; a *boundary* is a route the formulas themselves put at or above rho.

Routes



solid: exact on this curve dashed: construction family, not exercised dotted: existence open or heuristic

Ratio of the best carried route to rho, every curve



Units: $\log_2$ of estimated bit operations under the model in the report's `cost_model` block (Karatsuba multiplications, ten multiplications per elliptic operation, $10g^2$ per genus-g Jacobian operation, hyperelliptic index calculus at $g!\,q + g^2 q^{2-2/g}$, Enge–Gaudry by an explicit count of smooth polynomials for $g \le 1024$ and $L(1/2, \sqrt{2})$ above, Gaudry–Diem decomposition at the Bézout bound). Population: every registered curve with a carried route. No uncertainty interval applies: these are formulas, not measurements; the constants are the uncertainty, and changing one moves every point.

Findings

Registry SHA-256 `f8297bb8…4441fba`; 321 curves, 1156 routes. Verdicts: 314 boundary, 3 unresolved, 2 formula advantage, 2 formula marginal.

curve	route (genus / field)	rho	total	exp	ratio	class
c2pnb304w1	GHS m=9 upper (256 / $\mathbb{F}_4$)	160.2	102.7	+58.0	+57.6	unresolved
c2pnb272w1	GHS m=9 upper (256 / $\mathbb{F}_4$)	144.0	102.6	+42.0	+41.3	unresolved
c2pnb176w1	GHS m=8 lower (127 / $\mathbb{F}_4$)	95.0	71.1	+21.9	+23.9	unresolved
Oakley Group 4	GHS m=5 (15 / $\mathbb{F}_{2^{37}}$)	106.6	95.6	+14.4	+10.9	advantage
c2pnb368w1	GHS m=4 (7 / $\mathbb{F}_{2^{92}}$)	192.7	182.6	+18.3	+10.1	advantage
Oakley Group 3	GHS m=5 (15 / $\mathbb{F}_{2^{31}}$)	90.4	89.2	+5.4	+1.1	marginal
toy $\mathbb{F}_{p^5}$	cover descent (5 / $\mathbb{F}_p$)	32.0	31.6	−2.7	+0.4	marginal

1. **Every deployed prime-field curve is a boundary** on every enumerated route (best ratio −2.1 to −2.0 bits, the same-field genus-2 cover).
2. **Every Koblitz, subfield and prime-degree binary curve is a boundary** (Koblitz $m = 1$; random-b prime degree $m = n$).
3. **Two exact formula advantages**, both on deprecated or unused standards: Oakley Group 4 (genus 15 over $\mathbb{F}_{2^{37}}$, +10.9 bits) and c2pnb368w1 (genus 7 over $\mathbb{F}_{2^{92}}$, $q^{12/7} = 2^{157.7}$ against rho's 2^{176}, +10.1 bits).
4. **Three unresolved X9.62 w1 curves**: $\mathrm{Tr}(a) = 1$ with n even over $\mathbb{F}_4$ leaves the magic number at 8 or 9; one bound beats rho and the other does not. The verdict waits on the reduced a.

5. **Correction:** the earlier cover-route note priced the Oakley genus-16 cover with $q^{2-2/g}$ alone ($2^{17.7}$, 2^{22}). Gaudry's $g!\,q$ term dominates at this genus; the screen gives +1.1 and +10.9 bits. The census itself is reproduced exactly.

6. `ec_trapdoor::audit_curve` over-counts the magic number when a is not reduced (7–15 where the b-only span is 4–8 on the w1 curves); the screen prices the reduced value and records both.

7. The GHS isogeny extension is never the best carried route on a registered curve.

What is and is not established

- **Exact:** GHS magic numbers $\dim \mathrm{Span}\{(1, \sigma^i(\sqrt{b}))\}$, types and genera for every factorisation $N = n\ell$ of every binary curve where a reduces into $\mathbb{F}_2$ (when $\mathrm{Tr}(a) = 1$ and n is even, both bounds m_b, $m_b + 1$ are priced as heuristic rows); the degrees of the irreducible factors of $x^n - 1$ over $\mathbb{F}_2$ and the resulting census of $\sqrt{b'}$ with a given annihilator degree; same-field cover genera and degrees (from `COVERS.md`); whether $\#E$ is even.

- **Formula:** every cost, ratio and class.

- **Heuristic:** isogeny-extension rows assume the isogeny class is a random set of $\sqrt{|t^2 - 4q|}/\pi$ curves; none exhibits an isogenous curve.

- **Proposed:** base-change and cover-descent rows price the most favourable genus $g = k$; existence of such a cover for the curve is open.

- **Not done:** no cover is constructed, no map's action on the subgroup is verified, no relation is collected, nothing is timed. No speedup, key-recovery or deployed-curve security claim is made.

Automation

`jacobian_transfers build` rebuilds `docs/curves/jacobian-transfers.json` from the registry; `check` rebuilds with the committed parameters and diffs; `table --write-doc` refreshes the summary in `JACOBIAN_TRANSFERS.md`; `chart` regenerates the SVG and the TikZ above from the same data. The workflow `jacobian-transfers.yml` runs `check` on pull requests touching the registry or the screen, on pushes to `main`, weekly, and on dispatch, so a registered curve that has not been screened fails CI.